

COTTESMORE, United KingdomWMO: **034530**Lat: **52.736N**Lon: **0.649W**Elev: **461**StdP: **14.45**Time Zone: **0.00 (EUW)**Period: **94-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.2	26.8	20.4	15.6	28.5	22.7	17.3	30.2	34.5	48.1	31.3	47.9	7.0	270	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	79.0	63.8	75.1	62.7	71.8	61.1	65.7	74.4	64.1	71.9	62.6	69.6	12.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	86.9	68.8	61.1	81.8	66.5	59.7	77.7	65.2	30.7	74.3	29.5	72.2	28.4	69.7	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	26.1	23.4	19.5	84.8	3.4	2.5	17.0	86.7	15.0	88.2	13.1	89.6	10.6	91.4
			19.0	68.5	3.3	2.1	16.7	70.0	14.8	71.2	12.9	72.3	10.5	73.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	39.0	39.7	42.7	47.3	52.1	57.5	61.4	61.3	57.1	51.0	44.0	38.6
	DBStd	9.89	5.75	5.85	5.44	5.51	5.38	5.05	4.73	4.68	4.47	5.28	5.77	6.68
	HDD50	1621	343	290	233	114	38	2	0	0	3	50	191	357
	HDD65	5769	808	708	692	532	399	231	135	137	241	435	631	820
	CDD50	1387	1	2	6	32	105	228	353	352	215	81	10	2
	CDD65	58	0	0	0	0	0	8	23	23	3	1	0	0
	CDH74	431	0	0	0	1	6	64	169	164	22	5	0	0
	CDH80	62	0	0	0	0	0	8	23	30	1	0	0	0
(14) Wind	WSAvg	11.8	13.0	13.1	12.9	11.8	11.8	11.1	10.9	10.7	11.1	12.0	11.9	11.7
(15) Precipitation	PrecAvg	24.7	2.0	1.7	1.5	1.8	1.8	2.0	2.3	2.3	2.2	2.4	2.4	2.0
	PrecMax	31.4	3.4	3.2	3.0	5.2	4.3	5.2	4.3	6.0	4.7	4.6	4.1	4.1
	PrecMin	17.7	0.5	0.4	0.2	0.1	0.3	0.3	0.5	0.3	0.3	0.9	1.2	0.7
	PrecStd	4.3	0.8	0.7	0.8	1.3	1.0	1.3	1.2	1.4	1.3	1.0	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	55.2	59.6	68.7	73.6	80.8	83.3	83.9	77.5	68.0	58.8	55.2
		MCWB	50.9	51.1	50.8	56.1	59.6	65.7	65.3	65.1	62.0	59.6	55.9	52.7
	2%	DB	51.8	52.5	56.4	63.1	69.2	75.0	78.8	78.8	71.7	63.0	56.4	52.9
		MCWB	49.1	49.3	49.7	53.3	58.7	62.1	63.8	64.0	61.3	57.6	53.8	50.8
	5%	DB	49.8	50.3	53.8	59.8	65.4	70.8	74.7	73.9	67.6	60.6	54.2	50.4
		MCWB	47.5	47.1	48.6	51.2	56.4	60.4	62.8	62.6	59.0	56.5	51.7	48.4
	10%	DB	47.6	48.4	51.6	56.7	61.9	67.3	71.1	70.4	64.9	58.7	52.1	48.1
		MCWB	45.5	45.6	47.1	49.6	54.1	58.2	60.8	60.8	57.4	55.1	49.8	46.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	52.7	53.8	57.4	62.4	67.3	67.5	68.0	64.7	61.4	56.6	53.0
		MCDB	53.3	54.4	56.9	65.6	69.6	77.3	77.0	75.8	72.5	66.7	58.2	54.5
	2%	WB	49.5	50.0	51.2	54.6	59.6	63.5	65.4	65.7	62.5	59.3	54.2	51.1
		MCDB	51.3	52.2	54.6	61.1	67.4	71.8	74.6	73.8	68.8	62.1	56.0	52.8
	5%	WB	47.8	47.7	49.4	52.5	57.3	61.4	63.9	64.0	60.8	57.4	52.1	48.6
		MCDB	49.5	49.7	52.7	58.3	63.9	68.3	71.9	71.5	65.4	59.9	53.6	50.0
	10%	WB	45.6	45.9	47.8	50.4	55.1	59.6	62.2	62.4	59.1	55.8	50.3	46.5
		MCDB	47.4	48.0	50.9	55.6	60.7	65.9	69.5	68.9	63.2	58.2	51.8	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	10.0	12.2	14.5	14.9	15.4	15.8	15.5	13.6	11.4	9.5	8.3
		MCDBR	10.5	11.2	14.9	18.6	20.0	21.0	21.5	20.9	17.3	12.4	10.0	9.8
		MCWBR	9.9	9.2	9.7	10.6	11.4	10.7	9.3	8.7	8.6	8.4	8.8	9.5
	5% WB	MCDBR	10.3	10.8	13.2	17.1	18.6	18.6	18.8	18.0	15.4	11.8	9.9	9.8
		MCWBR	10.1	9.6	9.5	10.6	11.4	10.6	9.3	8.6	9.1	9.1	9.1	9.6
(40) Clear-Sky Solar Irradiance	taub		0.320	0.335	0.367	0.393	0.397	0.399	0.399	0.387	0.367	0.348	0.322	0.309
	taud		2.364	2.393	2.297	2.233	2.262	2.278	2.285	2.345	2.407	2.442	2.416	2.341
	Ebn at Noon		211	245	258	266	271	271	268	265	257	237	209	191
	Edh at Noon		19	24	33	39	40	40	39	35	29	23	18	16
(44) All-Sky Solar Radiation	RadAvg		233	453	792	1217	1508	1600	1544	1273	950	551	287	190
	RadStd		21	54	95	135	98	155	166	89	86	47	32	20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page